

Impedance assessment of a dual-resonance acoustic liner



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ARTICLE INFO

Article history:

Received 18 June 2014

Received in revised form 16 December 2014

Accepted 13 January 2015

Available online 5 February 2015

Keywords:

Acoustic liners

Absorption

Noise control

ABSTRACT

Acoustic liners are commonly used to reduce noise from commercial aircraft engines. Engine liners are placed in the nacelle inlet and aft bypass duct to attenuate the noise radiated from the engine. Traditional engine liners are constructed of a perforated facesheet over a honeycomb structure to create a quarter-wave absorber. With this design, the low frequency performance of the liner is limited by the depth of the honeycomb. However, with advances in engine design, lower frequency sound absorption is becoming more critical while liner depth must be minimized. Acoustic metamaterials can exhibit unique acoustic behavior using periodically arranged sub-wavelength resonators. Researchers have shown that acoustic metamaterials can effectively block the propagation of low-frequency acoustic waves. Therefore, acoustic metamaterial-inspired concepts are being investigated to improve the low frequency performance of engine liners. A proposed dual-resonance liner is presented here that combines the idea of a Helmholtz resonator metamaterial with a traditional quarter-wave acoustic liner. The low frequency acoustic absorption of a traditional liner can be significantly increased by adding a second, low frequency resonance to the system. The normal incidence absorption coefficient of the proposed liner is more than 10 times larger than a conventional honeycomb liner at the designed Helmholtz resonance frequency while retaining similar performance at higher frequencies.

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1. Introduction

In order to meet community noise standards, there is a need to reduce noise from turbofan aircraft engines. To combat the noise created by the engine, treatment is applied to the interior walls of the nacelle inlet and aft-fan ducts. Current treatments, or liners, attenuate the noise radiated from the engine with the greatest reduction occurring at the first harmonic of the blade pass frequency, which is commonly seen at around 1500 Hz. However, new engine and turbofan blade designs will require greater attenuation at lower frequencies than are achievable with the current liner design due to their depth restriction, while retaining comparable performance at higher frequencies. Therefore, new methods must be developed to achieve lower frequency noise reduction.

Historically, acoustic engine liners consist of a porous, metal or composite facesheet bonded on top of a honeycomb core that is fixed to a rigid backplate. Fig. 1 depicts a traditional perforate-over-honeycomb-core (POHC) acoustic liner. The honeycomb core forms a two dimensional array of cavities, which creates individual quarter-wave resonators. The parameter d defines the size of the

honeycomb cells. While the cells are necessary for the liner to be locally reacting, the size of the cell has little impact on the performance of the liner when d is less than a $\frac{1}{4}$ of a wavelength. The perforated facesheet placed on top of the cavities increases the losses in the device, which broadens the absorption peak in the frequency domain. The perforate panel also adds an inertial component, which reduces the resonant frequency of the device by a relatively small amount. The depth of the cavities is the primary variable controlling the resonant frequency. The low frequency performance of POHC liners is typically limited by the available cavity depth, D . The design of the POHC liners is similar to that of micro-perforated panel absorbers, which typically consist of a micro-perforate panel backed by an acoustic cavity [1]. However, the motion of the micro-perforate panel is usually utilized for increased absorption [2–4], while POHC liners utilize the honeycomb core to limit this motion and retain locally reacting impedance for grazing incidence sound.

The absorption of POHC liners at low frequencies is increased with a larger depth; however, an increase in liner depth requires a corresponding increase in the thickness of the engine nacelle, which significantly increases weight, drag, and fuel burn. Therefore, new designs for acoustic liners are needed to improve low frequency absorption and increase the bandwidth of performance

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Nomenclature

a	empirical constant	V	Helmholtz resonator volume
C_D	discharge coefficient	v_{rms}	root-mean-square velocity
c	speed of sound	W	Helmholtz resonator neck width
D	liner depth	α	absorption coefficient
d	perforate hole diameter	α_w	absorption coefficient for a wide duct
k	acoustic wavenumber	γ	ratio of heat capacities
L	Helmholtz resonator neck length	ε	end correction
L'	effective neck length	ζ	normalized acoustic impedance
m	mass	θ	normalized acoustic resistance
P_{ref}	reference pressure	μ	dynamic viscosity
Pr	Prandtl number	ρ	density of air
r	reflection coefficient	σ	perforate open area
S	Helmholtz resonator neck open area	χ	normalized acoustic reactance
SPL	sound pressure level	ω	angular frequency
t	facesheet thickness		

without changing the depth of the liner. Other absorbing designs have been shown to increase bandwidth and low frequency absorption through stacking Helmholtz resonators [5,6] or perforated panels [7–9], however these significantly increase the size of the system. One potential solution is to utilize a type of engineered heterogeneous material called an acoustic metamaterial.

Within the literature, a metamaterial is considered to be any engineered material built of individual elements of small conventional materials, usually periodic, that exhibits properties not found in nature. Acoustic metamaterials were first classified as such by Li and Chan [10], however acoustic analysis of periodic media and sonic crystals had been studied earlier [11–14]. Metamaterials can exhibit unique properties such as increased transmission loss due to stop-bands, negative effective density and/or stiffness, and negative refractive index. These novel properties are due to the periodic impedance mismatch of elements. By introducing mechanically-resonant inclusions within an elastic medium, stop-bands can be formed at low frequencies where the elastic wavelengths are much larger than the inclusion size [15]. Using the same methodology but with greater coupling to air, Hu et al. [16] developed an acoustically resonating sonic crystal using an array of split cylinder Helmholtz resonators. This implementation placed the resonant devices within the fluid medium. In contrast, Fang et al. [17] developed an ultrasonic metamaterial with Helmholtz resonators placed on the side of a waveguide. Further studies were performed by Cheng et al. [18,19] to investigate transmission versus the number of cells, periodic constants, and configuration of these one-dimensional metamaterials. Their studies showed significant transmission loss at the resonant frequency of the Helmholtz resonators. Similarly, García-Chocano et al. [20] demonstrated a large reduction in the transmitted waves in a duct using a quarter-wave resonator metamaterial, which was similar

to a traditional POHC liner. Therefore, it should be possible to incorporate these metamaterial structures into an acoustic liner to increase the low frequency performance.

The goal of the present work is to increase the low frequency noise reduction of an acoustic engine liner by utilizing acoustic metamaterials. Specifically, a new metamaterial-inspired liner is presented, which combines a traditional perforated panel over honeycomb liner with an array of Helmholtz resonators. To assess the noise reduction potential of an engine liner, the normal incidence impedance of the liner is determined before considering the grazing incidence aeroacoustic environment. Therefore, analytical normal incidence impedance models of a POHC liner, a Helmholtz resonator, and the proposed dual-resonance liner are presented. This is followed by a description of a finite element model of the liners. The experimental setup and procedure used to validate the models are then presented. Then, the results and analysis are given. Finally the conclusions are presented along with ideas for future work.

2. Analytical modeling

To predict the acoustic characteristics of candidate engine liners, the normal incidence, normalized specific acoustic impedance of a liner is estimated using a lumped-element analytical model. A model for a POHC liner is presented first followed by the analytical model of a single Helmholtz resonator. The final section describes the dual-resonance metamaterial-inspired acoustic liner and a lumped-element model for its normalized acoustic impedance.

2.1. Traditional POHC liner

Many researchers have proposed impedance models for a POHC liner. The model utilized in this work was described by Parrott and Jones [21]. The normalized specific acoustic impedance of the POHC liner can be described with four terms

$$\zeta_t = \theta + i\chi = \theta_{lin} + \theta_{nonlin} + i\{\chi_{fs} - \cot(kD)\} \quad (1)$$

where θ_{lin} and θ_{nonlin} are the linear and nonlinear resistance terms of the perforated facesheet, χ_{fs} is the reactance term of the facesheet, and the final term is the total impedance of the honeycomb core cavities. The impedance is normalized by the characteristic impedance of air, ρc . The impedance of the honeycomb core is purely reactive and is determined by the acoustic wavenumber k and the cavity depth D . The resistance of the perforated facesheet has both linear and nonlinear terms. These two terms can be described by a semi-empirical formulation of the flow resistance,

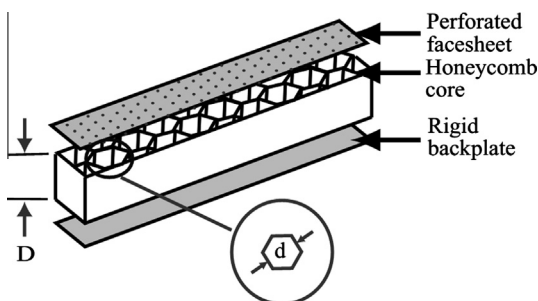


Fig. 1. Schematic of a perforate-over-honeycomb-core acoustic liner sample.

which is a function of the perforated facesheet parameters. The total normalized resistance is

$$\theta_f = \theta_{lin} + \theta_{nonlin} = \frac{a\mu t}{2\rho c(\sigma C_D)d^2} + \frac{\kappa_i + \kappa_e}{2c(\sigma C_D)^2} v_{rms} \quad (2)$$

where ρc is the characteristic impedance of air, μ is the ambient dynamic viscosity, t is the thickness of the facesheet, σ is the open area ratio, and d is the diameter of the perforated holes. There are two empirical terms [21]: a is assigned a value of 64 and the quantity $(\kappa_i + \kappa_e)$ accounts for the exit and entrance end effects of the holes and is set to unity. The discharge coefficient C_D represents the vorticity losses at the entrance and exit of the holes. A value of 0.771 is used in this model, which is based on orifice metering theory and the typical mass flow rates through a perforated facesheet with sharp edges [21]. The nonlinear term is a function of the root-mean-squared (RMS) acoustic particle velocity v_{rms} , which must be determined iteratively as it depends on the unknown impedance of the liner. The RMS velocity can be computed by

$$v_{rms} = \frac{P_{ref} 10^{SPL/20}}{\rho c |\zeta|} \quad (3)$$

where P_{ref} is the reference pressure (20 μ Pa), ζ is the impedance of the sample under test, and SPL is the sound pressure level at the surface of the facesheet. The solution to Eq. (3) is found by first guessing a value of v_{rms} in Eq. (2) for impedance, which will give a new value for v_{rms} to be used again in Eq. (2). This process is repeated until the solution of the impedance, Eq. (2), converges. The impedance will be computed for two different sound pressure levels and then compared to experimental data to assess the accuracy of the nonlinear resistance portion of the impedance model. It should be noted that the grazing flow effects described in Parrott and Jones [21] are neglected in this study.

The final term needed to determine the total normalized acoustic impedance is the reactance of the perforated facesheet. The reactance of the facesheet can be expressed as

$$\chi_{fs} = \frac{k(t + \varepsilon d)}{\sigma C_D} \quad (4)$$

where ε is the hole length end correction as defined by

$$\varepsilon = 0.85(1 - 0.7\sqrt{\sigma}) \quad (5)$$

The previous terms can be substituted into Eq. (1) to obtain the normalized impedance of the POHC liner.

2.2. Lumped element Helmholtz

Before the dual-resonance liner design and model is presented, a model of a single Helmholtz resonator is described. A simple

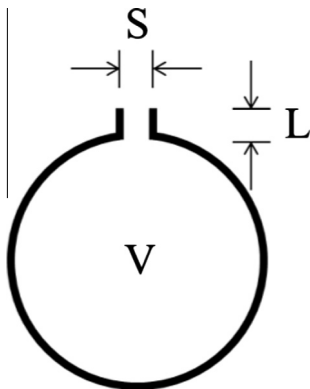


Fig. 2. Schematic of a Helmholtz resonator.

schematic of a Helmholtz resonator is shown in Fig. 2, where S is the area of the neck opening, L is the neck length, and V is the volume of the cavity. In the frequency range of interest, the resonator can be modeled as a mass-spring-damper system where the air within the neck is the mass, the fluid within the volume is the spring, and the acoustic and fluid losses are the damper. Thus, the normalized specific acoustic impedance of the Helmholtz resonator can be defined in terms of three components [22].

$$\zeta_h = \theta_w + i\{\chi_m - \chi_s\}/\rho c \quad (6)$$

where θ_w is the resistance due to the viscous wall losses in the neck, χ_m is the reactance due to the mass, and χ_s is the reactance due to the spring. The mass of the system is defined as

$$m = \rho S L' \quad (7)$$

where ρ is the density of air and L' is the effective length of the neck, which depends on the end conditions. For the Helmholtz resonator alone, the neck is in a fully flanged configuration, where $L' = L + 1.7W$ [23] and W is the width of the neck. The stiffness of the system is

$$s = \frac{\rho c^2 S^2}{V} \quad (8)$$

where c is the speed of sound of air within the volume of the resonator. The impedance of the mass and stiffness are defined as

$$\chi_m = \frac{\omega m}{S^2} \quad (9)$$

and

$$\chi_s = \frac{s}{\omega S^2} \quad (10)$$

respectively, where ω is the radial frequency, $2\pi f$.

The resistance of the Helmholtz resonator is due to the viscous wall losses in the neck and is calculated as

$$\theta_w = \frac{2mc\alpha_w}{S^2} \quad (11)$$

where the absorption coefficient for a wide duct is used [22,24]

$$\alpha_w = \frac{1}{Wc} \left(\frac{\mu\omega}{2\rho} \right)^{1/2} \left(1 + \frac{\gamma - 1}{\sqrt{Pr}} \right) \quad (12)$$

where W is the neck width, μ is the shear viscosity coefficient or dynamic viscosity, γ is the ratio of heat capacities, and Pr is the Prandtl number. It should be noted that because the neck of the resonator is short compared to the wavelength, it is assumed that the change in phase speed associated with the absorptive boundary layer in the neck does not significantly affect the reactance.

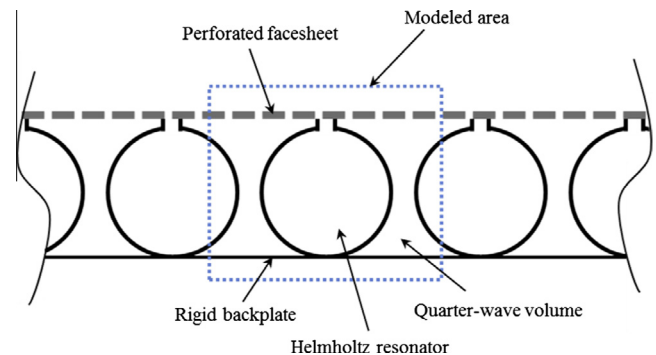


Fig. 3. Dual-resonance liner design schematic.

2.3. Dual-resonance liner design

The proposed dual-resonance metamaterial-inspired acoustic liner is designed as an array of Helmholtz resonators with a perforated facesheet over the necks, as seen in Fig. 3. Each of the individual Helmholtz resonators is separated by an open volume that behaves like the quarter-wave resonators within a traditional POHC liner. In this configuration, the walls of the Helmholtz resonators act as the structural supports for the perforated facesheet as well as the partitions to ensure the liner is locally reacting in a grazing incidence configuration. An advantage of this dual resonance design is the ability to tune the Helmholtz resonators to target certain frequencies by varying the length and width of the neck without affecting the depth of the liner, while retaining the quarter-wave resonance of the POHC liner. This design is similar to the MPP absorber design outlined by Lee et al. [25], which incorporates the Helmholtz resonator below the perforated facesheet and separates the acoustic volume into two separate regions.

The impedance of the dual-resonance liner is modeled with the perforated facesheet to investigate the combined impedance of the quarter-wave and Helmholtz resonators. To match the experimental procedure, the area outlined by the dotted line in Fig. 3 was analytically modeled. When the Helmholtz resonator neck and quarter-wave resonators are covered by the perforated plate in the dual-resonance design, the impedance of the Helmholtz resonator becomes

$$\zeta_{hp} = \zeta_h + \theta_{lin} + \theta_{nonlin} + i\chi_{fs} \quad (13)$$

which includes the resistance of the facesheet. It should be noted that in this case, the neck of the Helmholtz resonator is assumed to be unflanged and therefore the effective length is calculated as, $L' = L + 1.4W$. The perforated facesheet impedance terms are the same as presented in Section 2.1. Because the quarter-wave and Helmholtz resonators are parallel to each other and act as two branches at the end of a duct, the total impedance is found in the same manner as adding springs in parallel. However, because each component does not have the same radiating surface area, the impedances must be scaled by the ratio of areas. Therefore the total impedance of the dual-resonance liner design is computed as

$$\zeta^{-1} = \left(\frac{\zeta_t}{R_t} \right)^{-1} + \left(\frac{\zeta_{hp}}{R_h} \right)^{-1} \quad (14)$$

where R_t and R_h are the surface area ratios of the POHC and Helmholtz resonator components normalized by the total area of the modeled section.

3. Numerical modeling

In addition to the analytical model described above, finite-element numerical models of the liner samples were also created. The finite-element models included additional dynamics such as the compliance of the Helmholtz resonator walls and enabled direct calculation of the RMS velocity through the perforated facesheet without requiring the iterative approach needed with the analytical model. The finite-element models were created using a commercially available multiphysics finite-element modeling program [COMSOL Multiphysics™].

Three-dimensional finite element models of the liner samples were created to mimic the experimental setup in the normal incidence tube. Fig. 4 shows a model of the dual-resonance liner as an example. Specifically, a 50.8 mm square duct was modeled to represent the normal incidence tube used during testing. The 38.18 mm thick liner samples were included at one end of the tube opposite the source. Hard wall boundary conditions, zero normal acoustic velocity, were defined for the duct walls and honeycomb

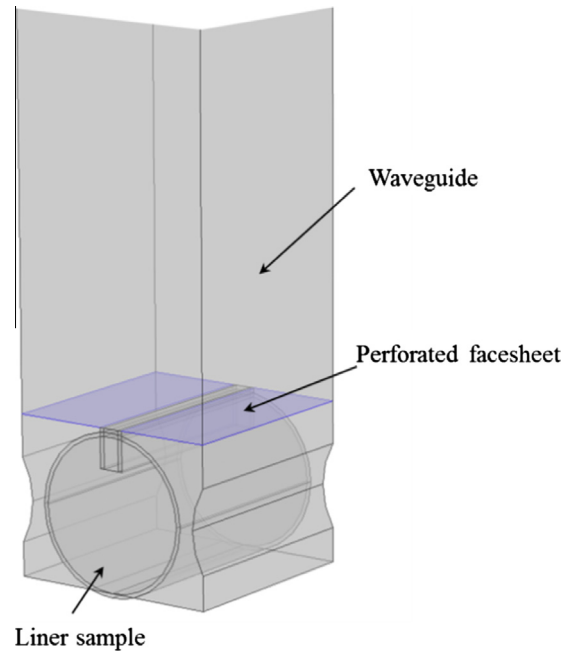


Fig. 4. A three-dimensional finite element model of the dual-resonance liner.

core. While the non-linear flow resistance of the perforated facesheets was calculated using Eq. (2), the impedance was determined using a built-in interior perforated plate model within the finite-element modeling program [26]. Recall that the analytical model required an iterative approach to solve for the value of v_{rms} . This step is not required for the finite element model since the impedance and RMS velocity at the facesheet are determined simultaneously and included in the impedance model directly. A single Helmholtz resonator was also modeled, with the elastic compliance of the sidewalls and neck included. Viscous losses in the neck of the Helmholtz resonator were included using a boundary-layer absorption pressure acoustics domain. All other domains were modeled as linear, lossless acoustic domains. The proposed dual-resonance liner model combined the compliant-walled Helmholtz resonator and an interior perforated plate model of the POHC. The impedance of each type of liner was found in the same manner as the experimental procedure outlined in the following section.

4. Experimental setup and procedure

Experimental validation of the impedance models was performed using the two-microphone method (TMM) [27] in NASA

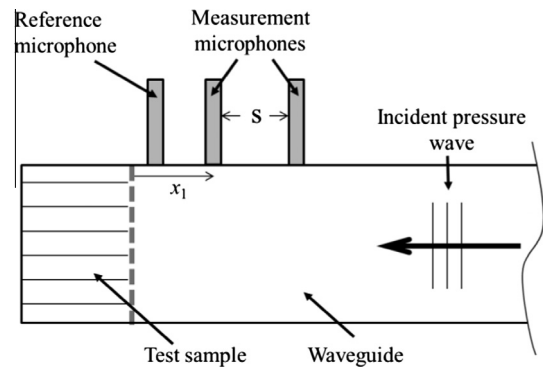


Fig. 5. NIT schematic.

Langley Research Center's normal incidence tube (NIT). A schematic of the NIT is shown in Fig. 5. Several test samples were evaluated, both with and without the perforated facesheet. Since the facesheets behave nonlinearly with amplitude, the samples with face sheets were tested at two different sound pressure levels, 120 dB and 140 dB, as measured by a reference microphone adjacent to the facesheet at a distance of 6.35 mm from the facesheet surface. Note that it was necessary to adjust the source level for each sample to obtain a consistent sound pressure level at the reference microphone. The TMM was then used to calculate the impedance of the samples. Specifically, the real θ and imaginary χ components of impedance were found versus frequency.

A convenient, but limited, metric to compare the *in situ* performance of acoustic liners is the absorption coefficient, α . This parameter is based on the reflection coefficient, r ; which is defined as

$$r = \frac{\zeta - 1}{\zeta + 1} \quad (15)$$

where ζ is the impedance of a given sample. The absorption coefficient was calculated from the experimental measurements as

$$\alpha = 1 - |r|^2 \quad (16)$$

4.1. Test samples

Test samples of the POHC liner, a Helmholtz resonator, and the dual-resonance liner were tested to determine the acoustic impedances. Fig. 6 shows a sample of a POHC liner while Fig. 7 shows a sample of the proposed dual-resonance acoustic metamaterial-inspired liner that was built using stereolithography. Although the latter sample is shown without the perforated facesheet, it was tested with a facesheet similar to the POHC liner. To validate the Helmholtz resonator impedance model, the quarter-wave cavities of the dual-resonance sample were filled in with clay, as shown in Fig. 8. The characteristics of the POHC liner are found in Table 1 and the Helmholtz resonator geometry is given in Table 2. Each sample has an active area subjected to the incident pressure wave of 50.1 mm by 50.1 mm.

5. Results and analysis

The real and imaginary parts of the predicted impedance are compared to the experimentally obtained values for the POHC liner, Helmholtz resonator, and the dual-resonance liner. Recall that samples with a perforated facesheet were tested at two sound pressure levels, 120 dB and 140 dB. The samples without a



Fig. 6. POHC liner sample.



Fig. 7. Dual-resonance liner sample without the perforated facesheet.

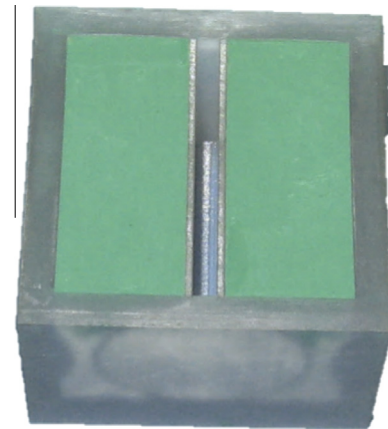


Fig. 8. Helmholtz resonator sample.

Table 1

Geometric properties of the perforate-over-honeycomb-core (POHC) liner.

Property	Value
Hole diameter (d)	0.965 mm
Facesheet thickness (t)	0.813 mm
Cavity depth (D)	38.18 mm
Open area ratio (σ)	0.089

Table 2

Geometric properties of the Helmholtz resonator.

Property	Value (mm)
Neck width (W)	4
Neck length (L)	2
Resonator diameter (d_{res})	36.3

facesheet were only tested at 120 dB. In the following figures, the real and imaginary components of impedance are shown versus frequency and are expressed as resistance θ and reactance χ , respectively. Lastly, the absorption coefficient of the dual-resonance liner is compared to the POHC liner at both sound pressure levels.

5.1. POHC impedance

The analytical, numerical and experimental impedance curves for a POHC liner sample are shown in Fig. 9. The nonlinear resis-

tance of the POHC has been captured by both modeling techniques, as demonstrated by the large change in resistance between the two sound pressure levels. Similarly, both models show good agreement to the experimental impedance for both resistance and reactance.

5.2. Helmholtz impedance

Fig. 10 shows the impedance comparison for the Helmholtz resonator alone. Without the perforated facesheet, the target resonant frequency of the Helmholtz resonator is 1200 Hz, which is in good agreement with the numerical and experimental data. Again, both the analytical and numerical models can predict the real and imaginary components of impedance. While both models accurately capture the trend in resistance versus frequency, the analytical prediction is slightly more accurate than the numerical prediction. For the reactance, the analytically and numerically computed values match the experimental data very closely. It should be noted that the reactance behavior is quite similar to the reactance of the POHC liner.

5.3. Dual-resonance liner impedance

Fig. 11 shows the impedance of the dual-resonance liner. In this case, the analytical model does not accurately predict the resistance or reactance of the liner. This disagreement most likely comes from the iterative solution of the RMS velocity calculation within the impedance model. The iterative approach cannot accurately converge when both the Helmholtz and quarter-wave resonances are affected by the nonlinear resistance of the perforated facesheet. In contrast, the predictions of the numerical model are a better match to the experimental data. The numerical model accurately predicts the change in impedance for the two sound pressure levels. At the low frequency resistance peak near 800 Hz, the numerical model under predicts the resistance at the 120 dB source level. However, all other frequencies show good

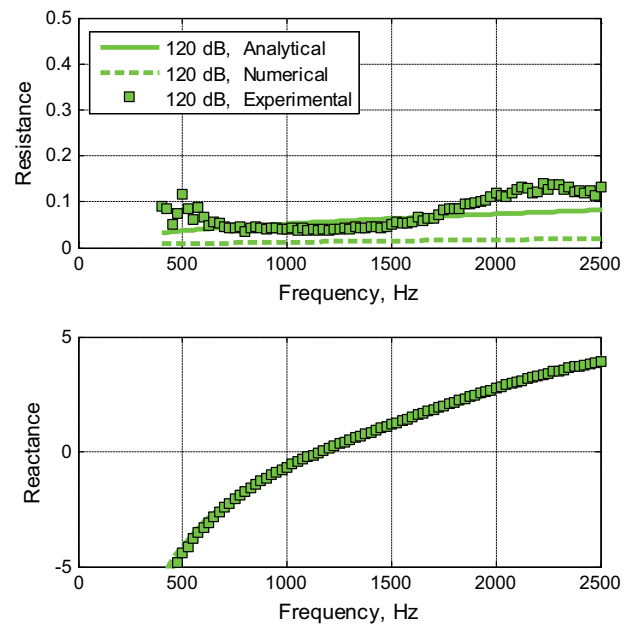


Fig. 10. Analytical, numerical, and experimental impedance of a Helmholtz resonator.

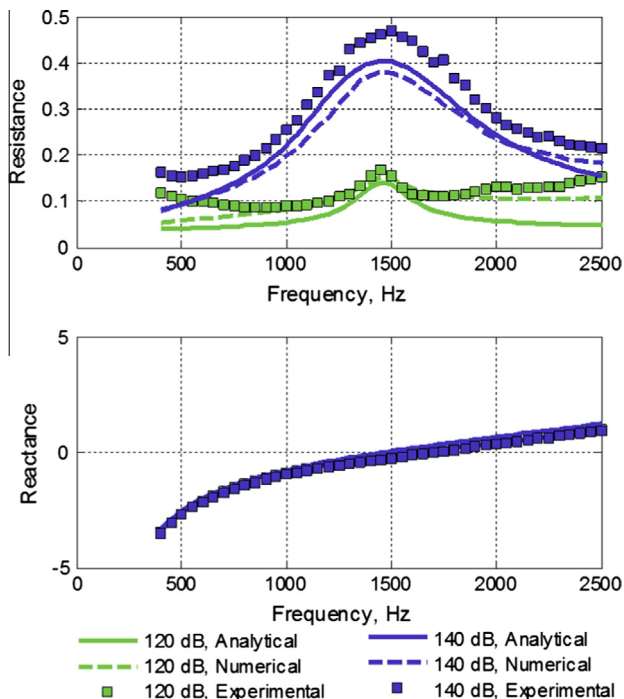


Fig. 9. Analytical, numerical, and experimental impedance of a perforate over honeycomb core liner for two sound pressure levels, 120 dB and 140 dB.

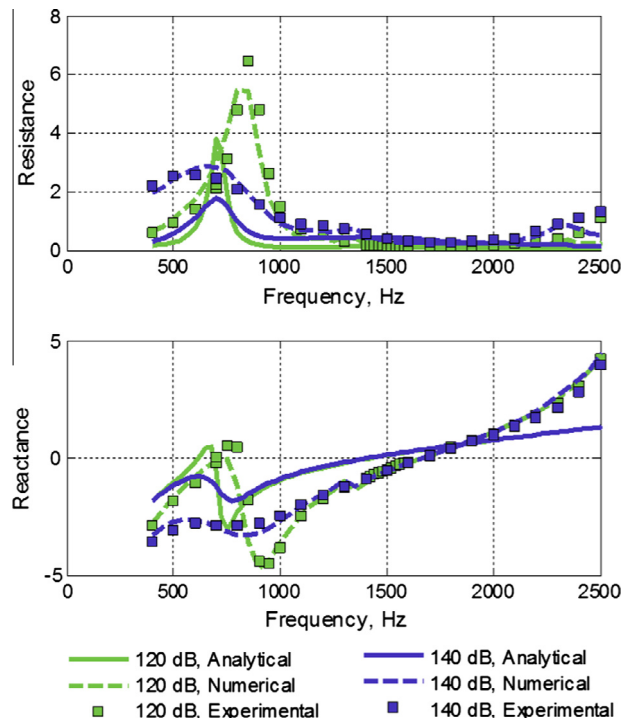


Fig. 11. Analytical, numerical, and experimental impedance of the dual-resonance liner concept for two sound pressure levels, 120 dB and 140 dB.

agreement between the numerical predictions and experimental data. It should be noted that in contrast with typical POHC liners, near resonance the resistance of the dual-resonance liner decreases as the acoustic level increases.

5.4. Absorption

The absorption coefficient of the dual-resonance liner is compared to the traditional POHC liner to quantify the performance

of the concept. As stated earlier, while convenient, the absorption coefficient is not sufficient to fully capture the liner benefit; the acoustic benefit must also be robust to a grazing incidence sound field as well as a grazing flow. The numerical and experimental absorption coefficient spectra of both liners for the 120 dB and 140 dB sound pressure levels are shown in Figs. 12 and 13, respectively. For both source levels, the absorption coefficients of both the POHC and dual-resonance liners are accurately predicted by the numerical model. Comparing the absorption between the two liners in Fig. 12, the dual-resonance liner shows a large increase between 400 and 1000 Hz over the POHC liner, with a maximum increase of 0.7 at 700 Hz. At frequencies above 1000 Hz, both liners have a similar response. In other words, the dual-resonance liner retains nearly the same high frequency absorbing performance of the POHC liner with a significant increase below 1000 Hz. The sharp peak occurring in the numerical model at 1300 Hz is due to erroneous deformation of the Helmholtz resonator neck, where the experimental resonator is constrained by the perforated facesheet. Similarly, there is a small under-prediction of absorption above 1400, due to the model under-predicting the acoustic velocity through the facesheet, which results in smaller values of absorption. For the 140 dB source level, shown in Fig. 13, the benefits of the dual-resonance liner are not as significant, however the proposed liner does increase absorption from 400 to 1000 Hz. Unfortunately, there is also a small absorption penalty above

1000 Hz. In summary, the dual-resonance liner shows an increase in absorption below 1000 Hz due to the presence of the Helmholtz resonator relative to the POHC liner alone.

There are two items that should be noted about the performance of dual-resonance liner:

- There is little interaction between the Helmholtz resonator and the quarter-wave resonator because their target resonance frequencies are tuned to significantly different frequencies, 800 and 1500 Hz respectively. The small interaction between the two does shift the Helmholtz resonance to slightly lower frequency and the quarter-wave resonance to slightly higher frequency.
- The volume under the perforated facesheet does not dictate the resonance, only the depth of the cavity and the area of the facesheet. Therefore, the volume can be reduced (up to a point) and still retain strong absorption. Because the Helmholtz resonator only reduces the area slightly and still allows for the quarter-wave resonator to be locally reacting, the addition of the Helmholtz resonator has only a small impact on the absorptive response.

6. Conclusion

Analytical and numerical models of a metamaterial-inspired acoustic liner were developed and verified experimentally. It was found that the finite-element numerical model more accurately predicted the impedance of the proposed dual-resonance liner design than the analytical model. The impedance of the dual-resonance liner was compared to that of a traditional perforate-over-honeycomb-core liner. It was found that the dual-resonance liner increases the absorption below 1000 Hz relative to the POHC liner, due to the Helmholtz resonance, with little impact at the quarter-wave resonance of the POHC. Future work will investigate the effects of the various resonator parameters on the impedance of the dual-resonance liner and will consider variations to the shape of the Helmholtz resonator volume.

Acknowledgement

Carol Harrison of the Structural Acoustics Branch at NASA Langley Research Center is acknowledged for her time and effort in the experimental testing of the liner samples.

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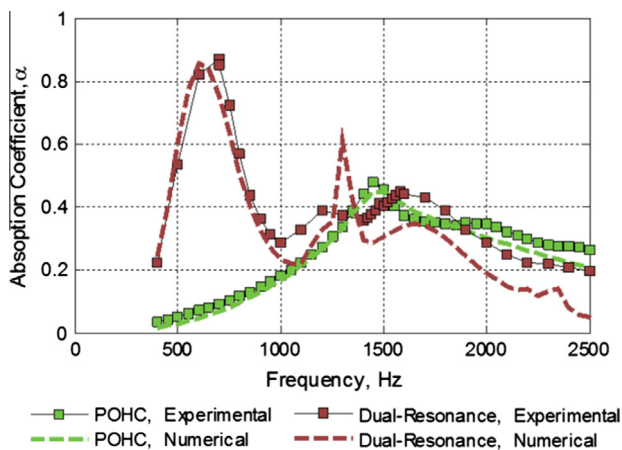


Fig. 12. Comparison of absorption coefficient, α , between the POHC liner and the dual-resonance liner for a 120 dB sound pressure level.

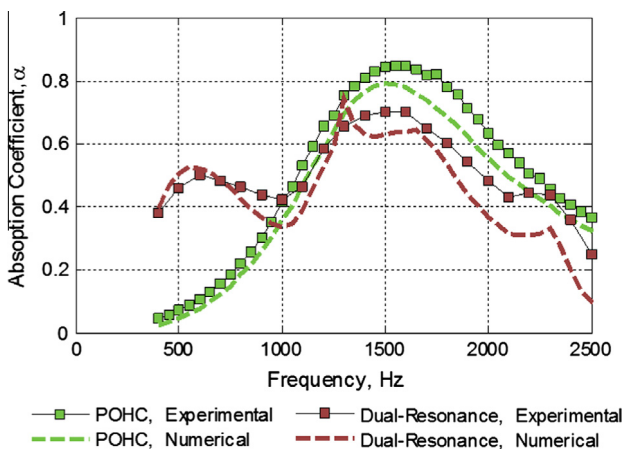


Fig. 13. Comparison of absorption coefficient, α , between the POHC liner and the dual-resonance liner for a 140 dB sound pressure level.

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